# FREEDOM INTERNATIONAL SCHOOL

## WORKSHEET

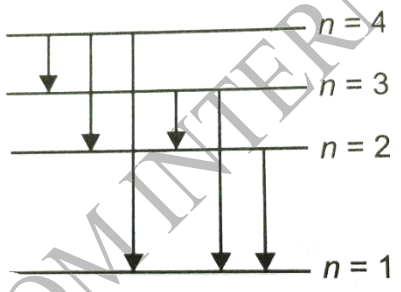
## CLASS XII

## PHYSICS

## ATOMS

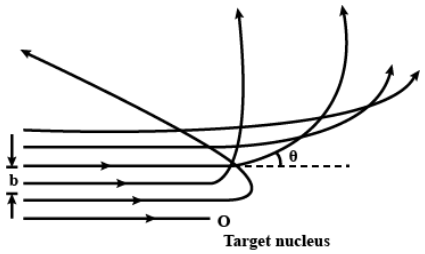
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1. In the Gieger- Marsden experiment, an α- particle of 5.12 MeV energy approaches a gold target (Z= 79), comes momentarily to rest and then reverses its direction. Find the distance of closest approach of α- particle to the target nucleus.
2. What is the angular momentum of an electron in the third orbit of an atom?
3. The ground state energy of hydrogen atom is -13.6 eV.
   (i) What is the kinetic energy of an electron in the $2^{nd}$ excited state?
   (ii) What is the potential energy of an electron in the $3^{rd}$ excited state?
   (iii) If the electron jumps to the ground state from the $3^{rd}$ excited state, calculate the wavelength of the photon emitted.
4. A monochromatic radiation of wavelength 975 A˚ excites the hydrogen atom from its ground state to a higher state. How many different spectral lines are possible in the resulting spectrum? Which transition corresponds to the longest wavelength amongst them?
5. The given figure shows the energy level diagram of a hydrogen atom.




   (i) Find out the transition which results in the emission of a photon of wavelength 496 nm.
   (ii) Which transition corresponds to the emission of radiation of maximum wavelength? Justify your answer.
6. Calculate the de- Broglie wavelength associated with the electron revolving in the first excited state of hydrogen atom. The ground state energy of a hydrogen atom is -13.6 eV.
7. The ground state energy of a hydrogen atom is -13.6 eV. If an electron makes a transition from an energy level -1.51 eV to -3.4 eV, calculate the wavelength of the spectral line emitted and name the series of hydrogen spectrum to which it belongs.
8. A 12.5 eV electron beam is used to excite a gaseous hydrogen atom at room temperature. Determine the wavelengths and the corresponding series of the lines emitted.
9. The trajectories, traced by different α- particles, in Geiger- Marsden experiment were observed as shown in the figure.

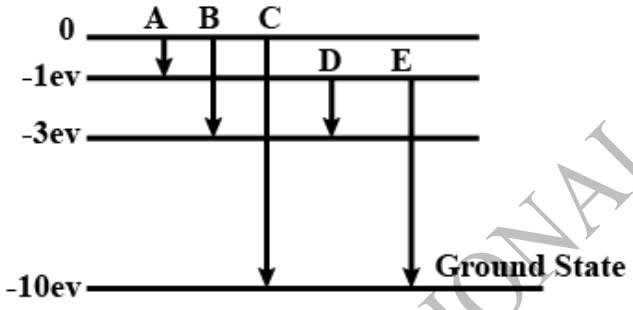
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What names are given to the symbols 'b' and 'θ' shown here?

What can we say about the values of 'b' for (i) $\theta \simeq 0^{\circ}$ and (ii) $\theta \simeq \pi$ radians?

10. The energy levels of an atom of element X are shown in the diagram. Which one of the level transitions will result in the emission of photons of wavelength 620 nm? Support your answer with mathematical calculations.




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